

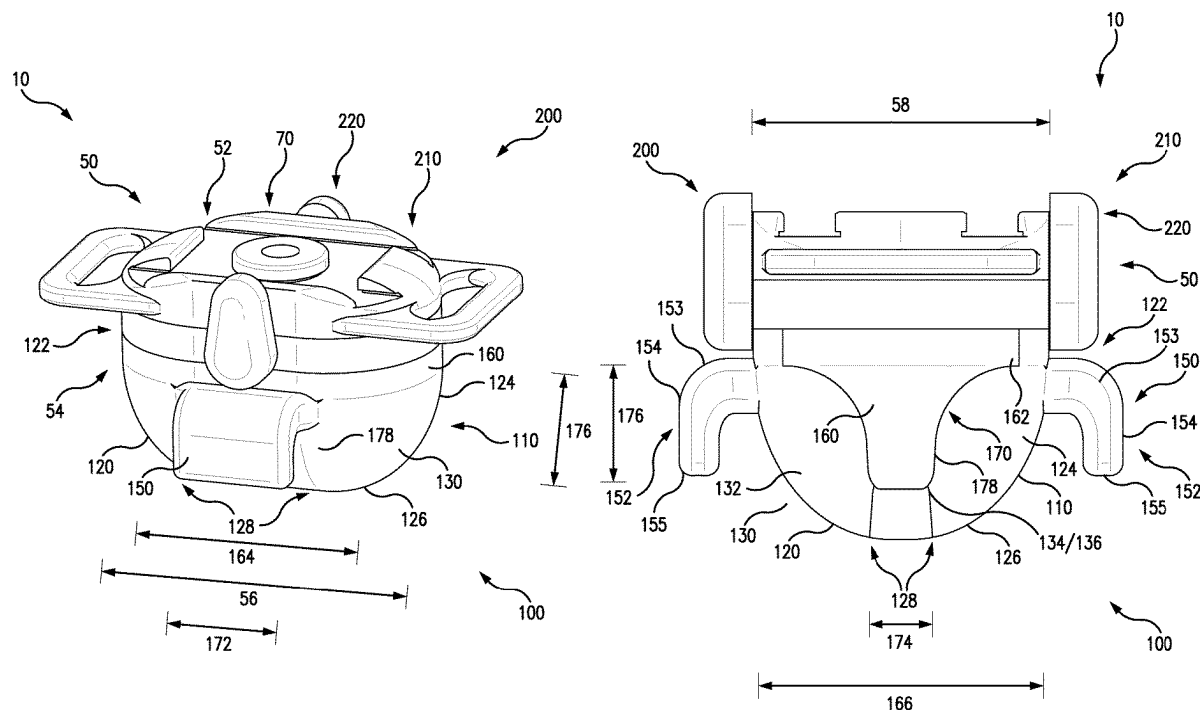
(45) **Date of Patent:** Sep. 2, 2025

2017/12004 (2013.01)

See application file for complete search history.

A hemostasis aiding device for use at wound sites associated with venous access procedures.

A61B 17/12 (2006.01)



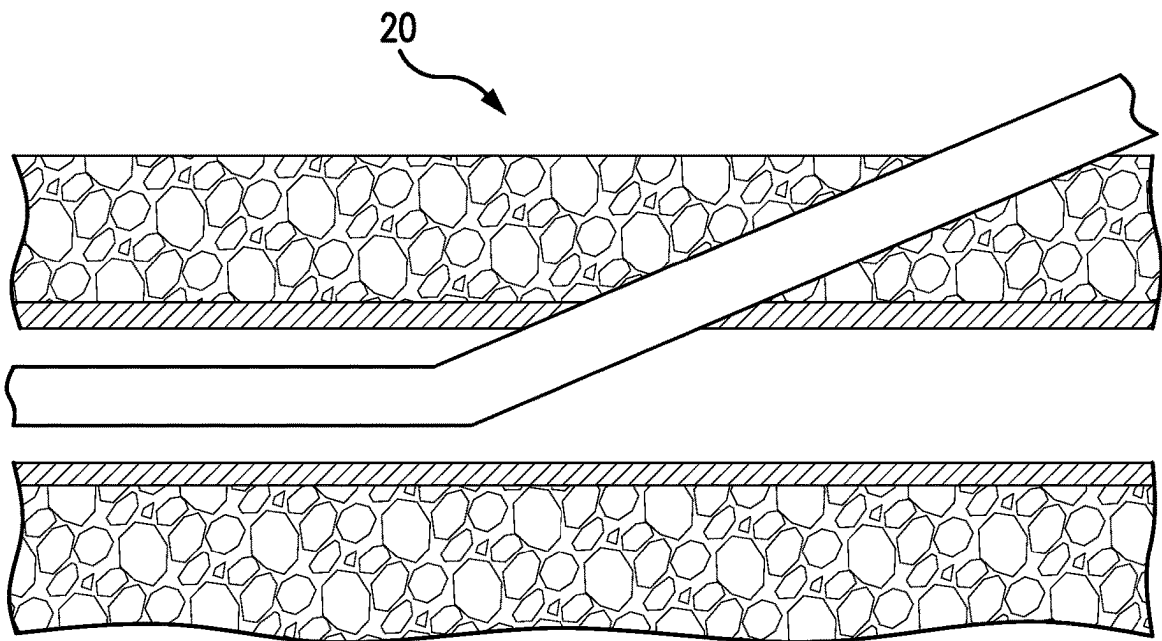


FIG. 1

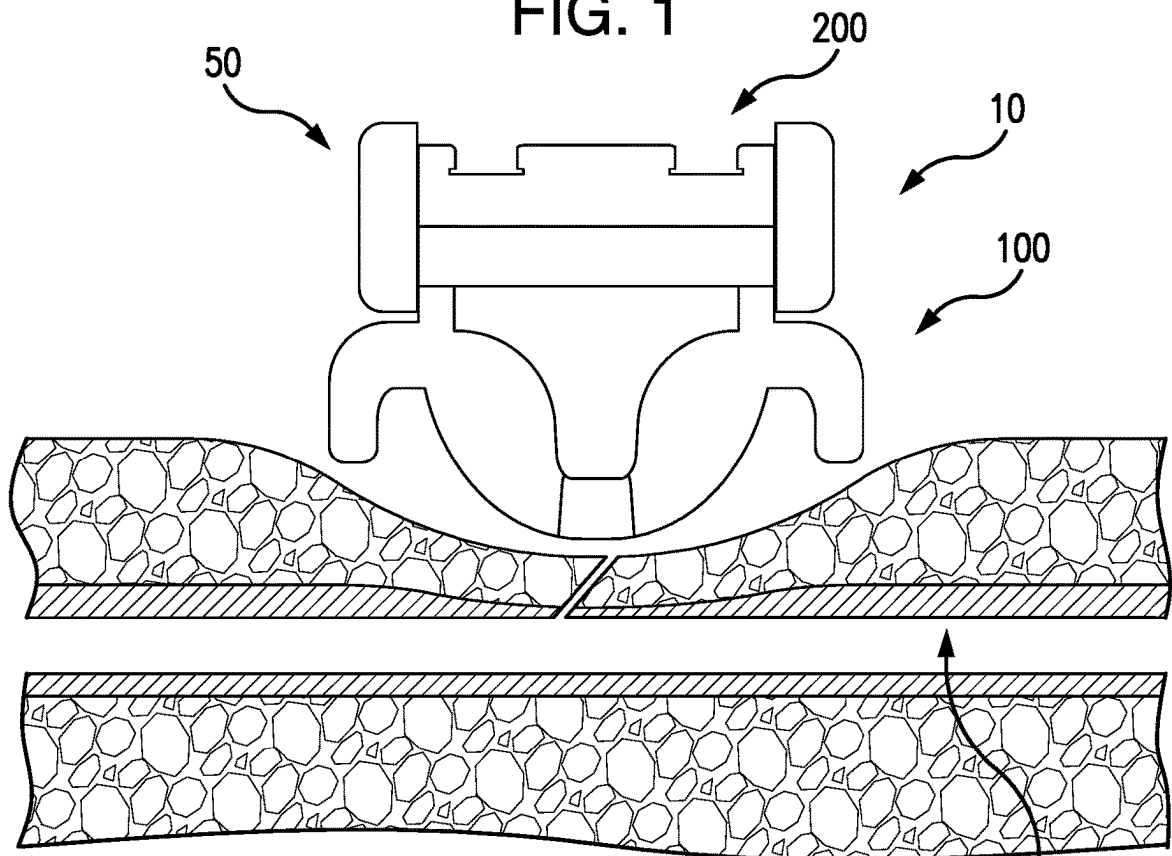


FIG. 2

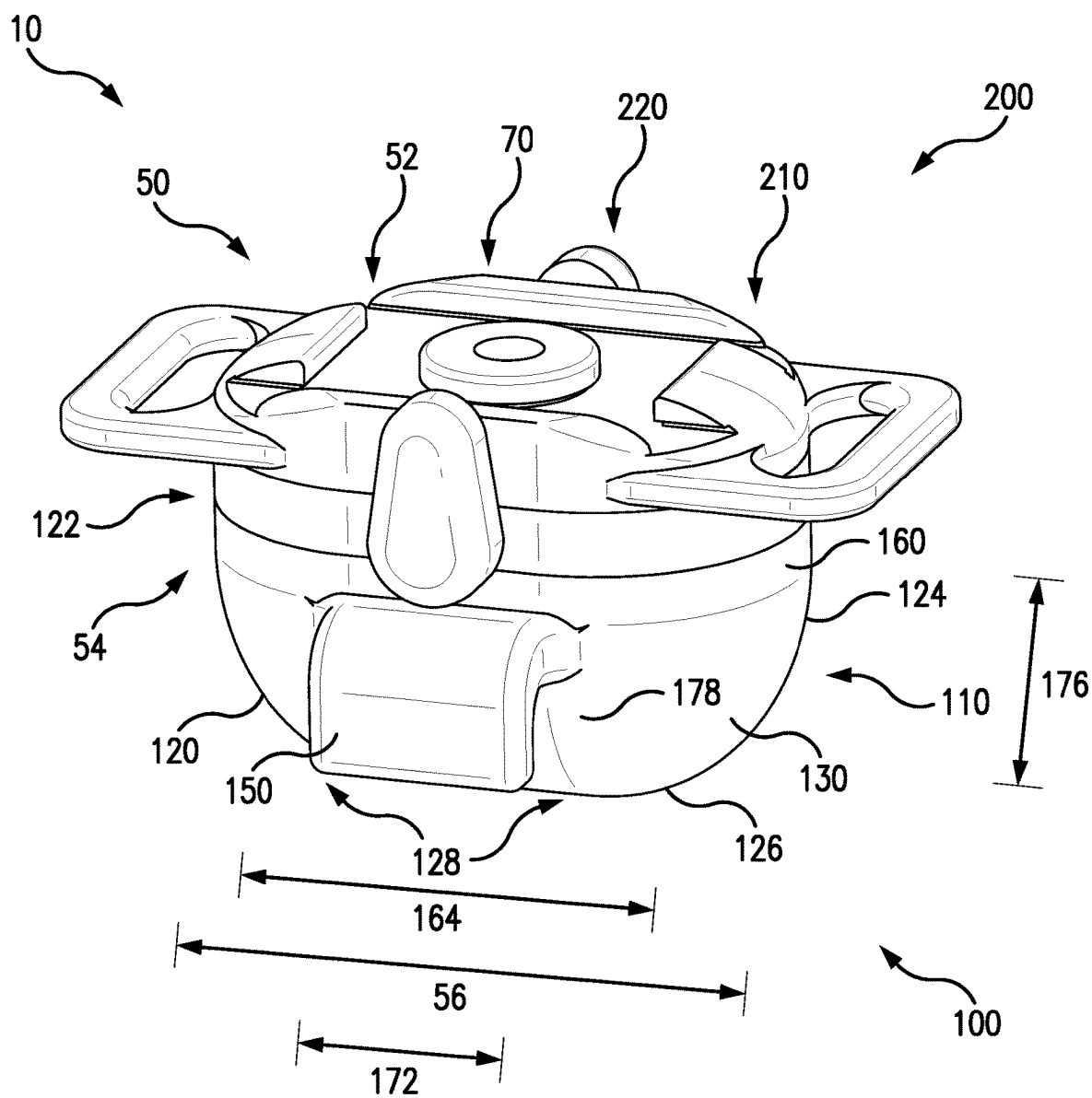


FIG. 3

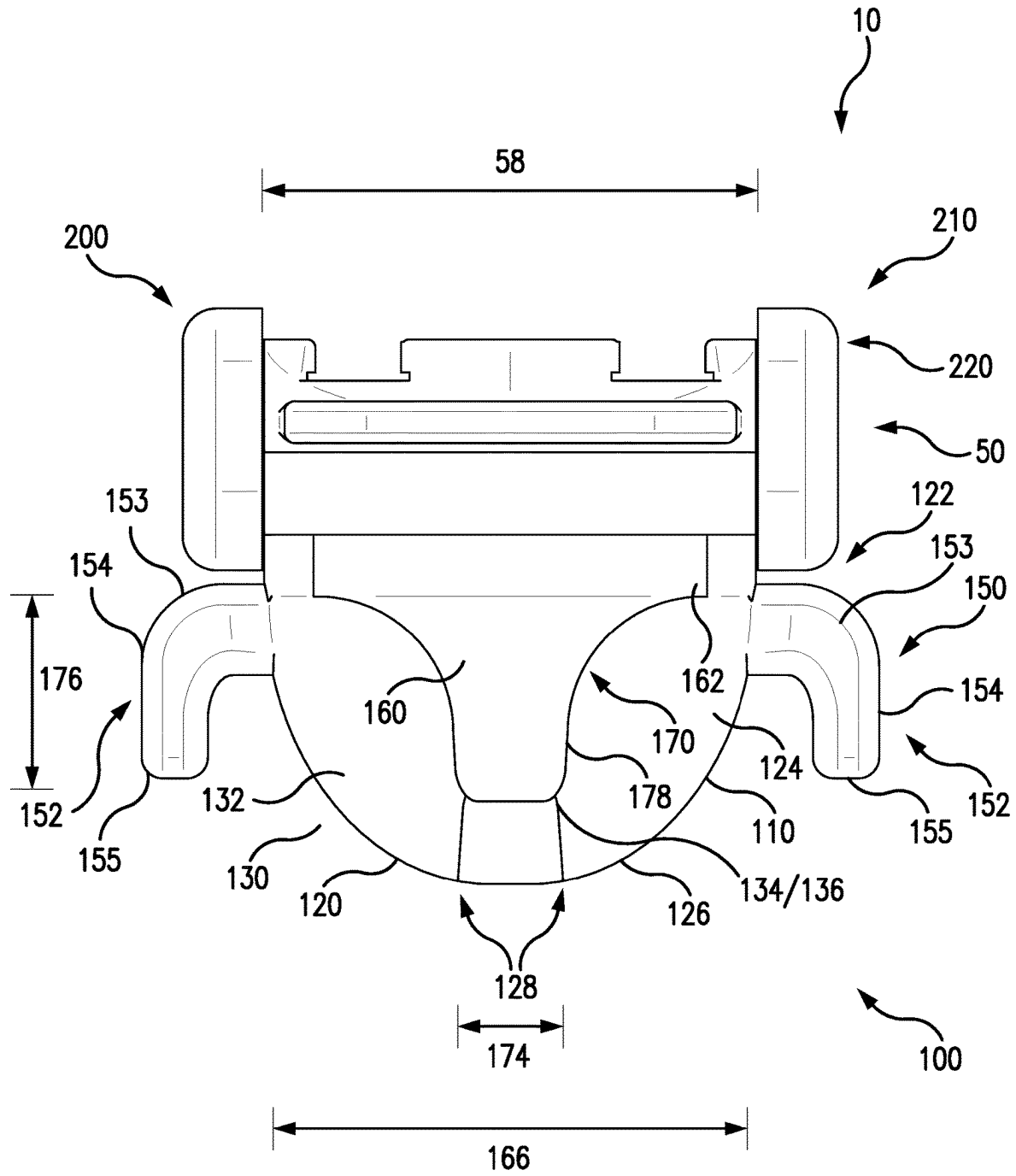


FIG. 4

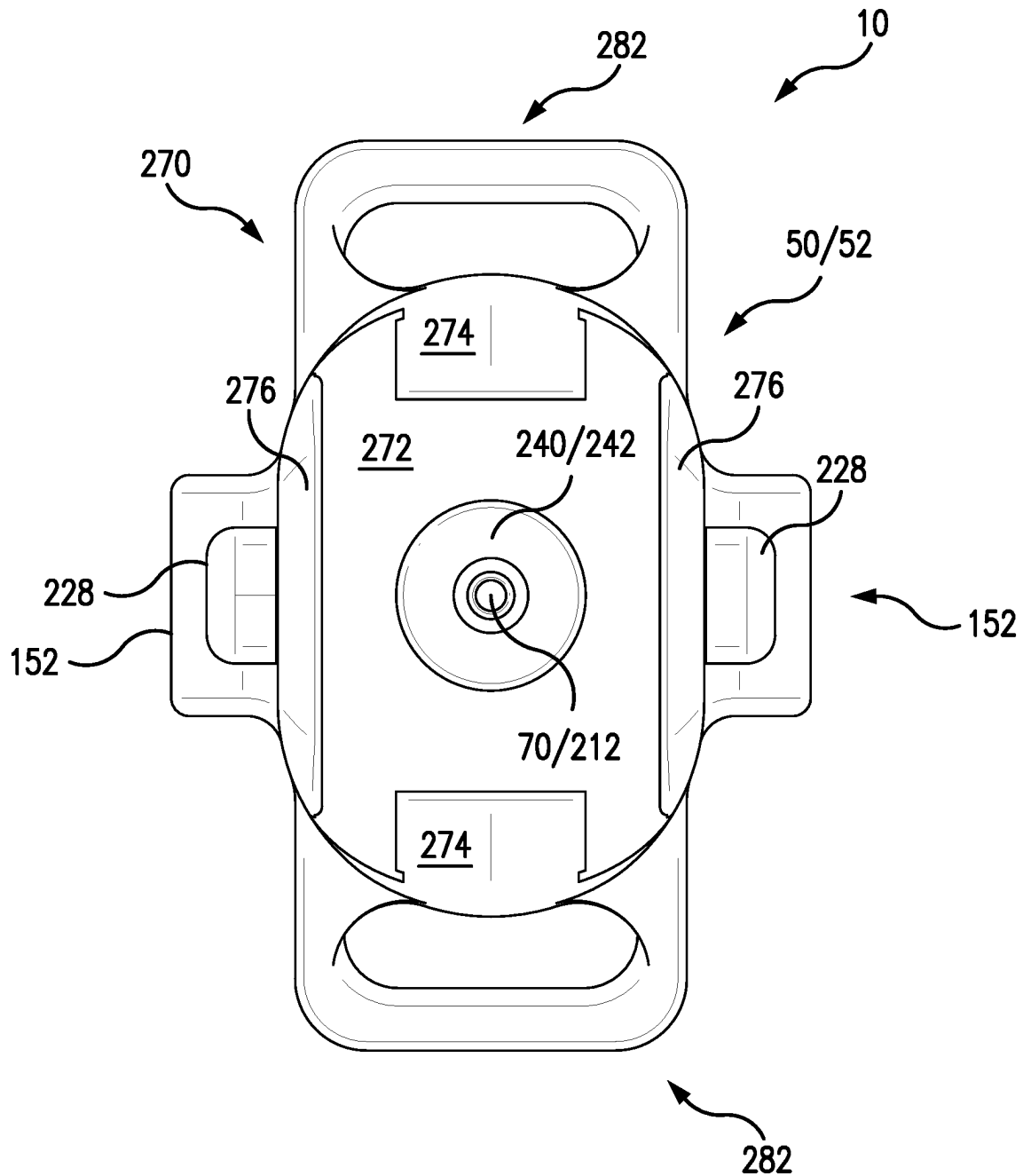
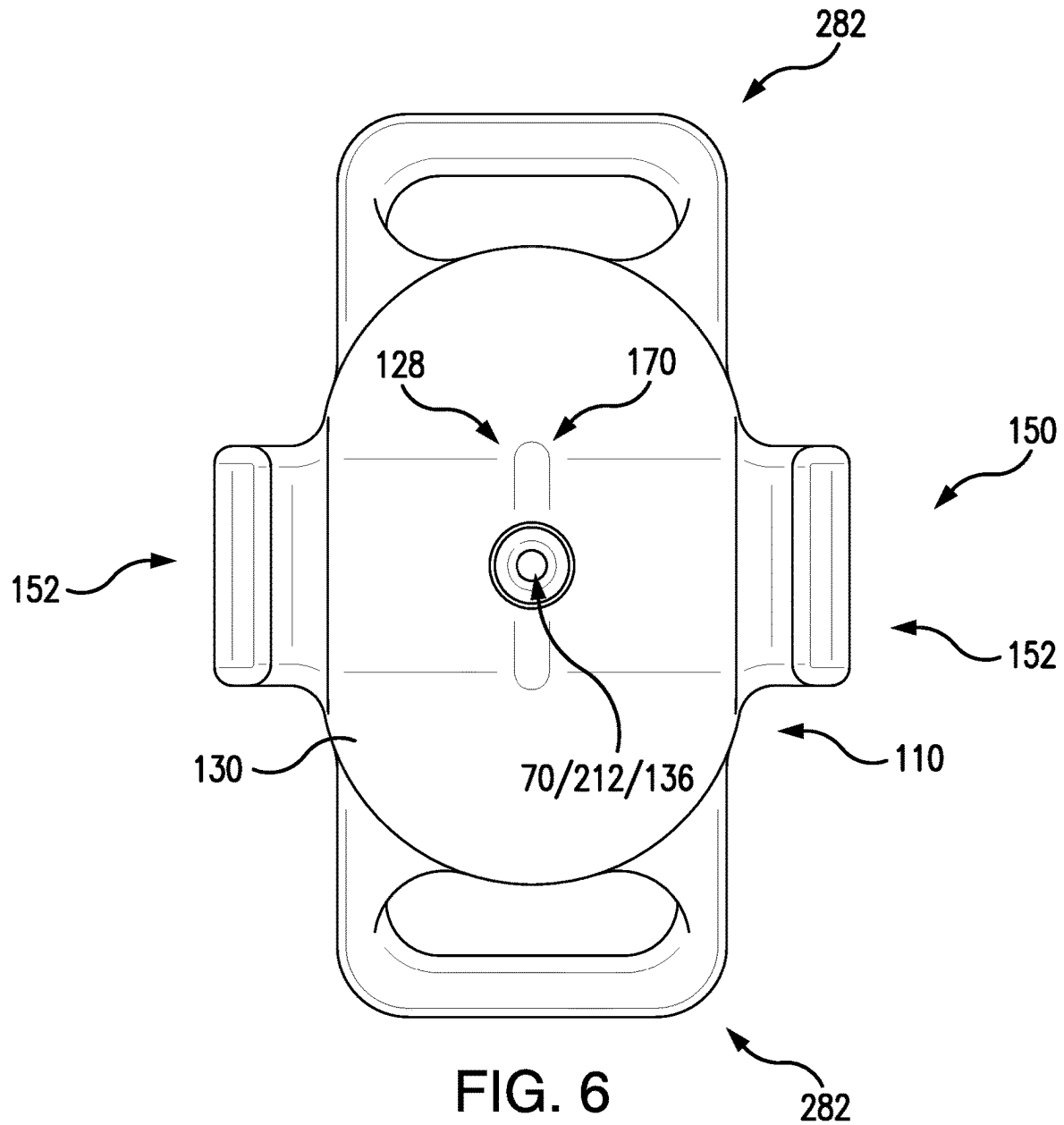


FIG. 5



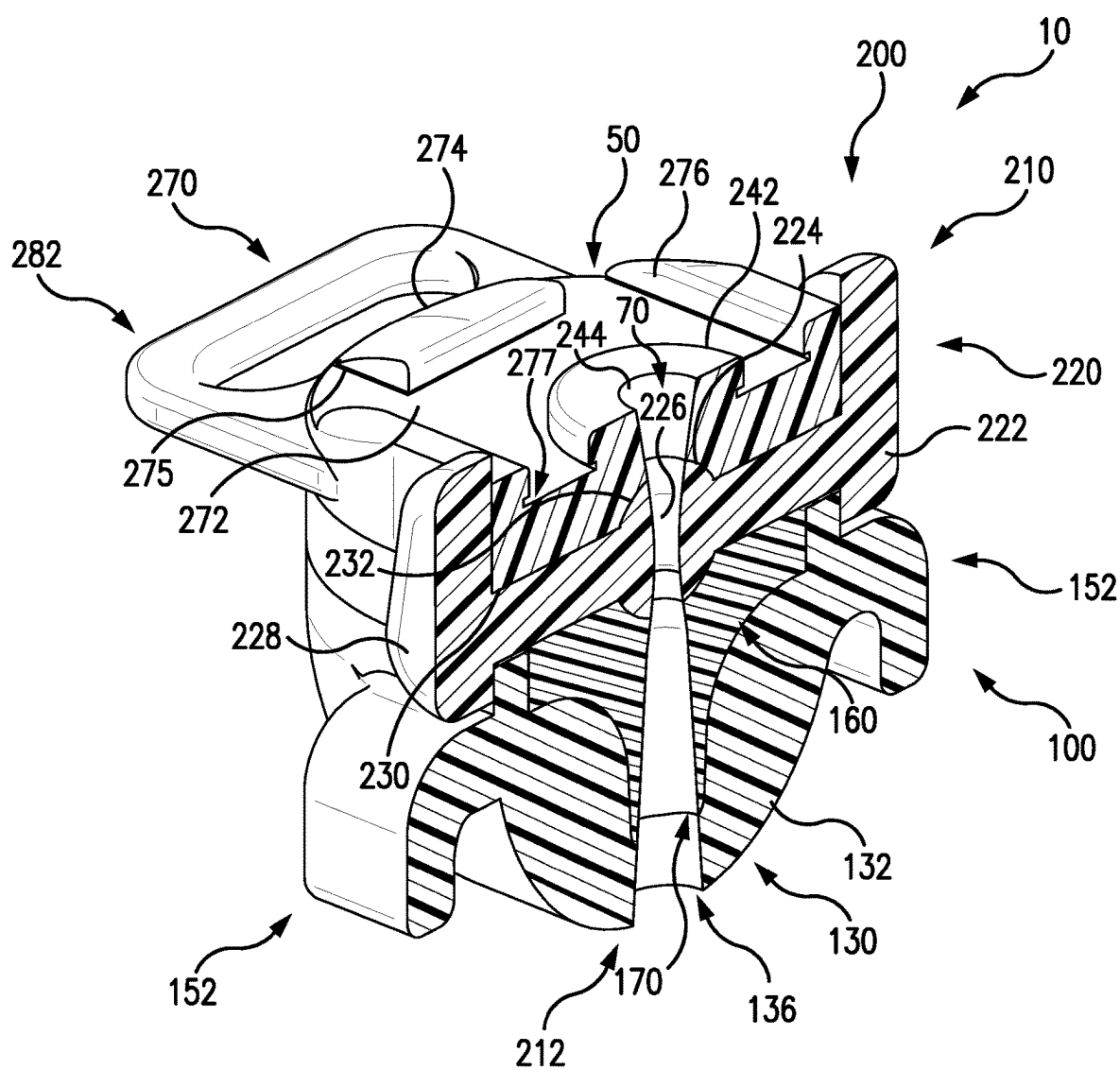


FIG. 7

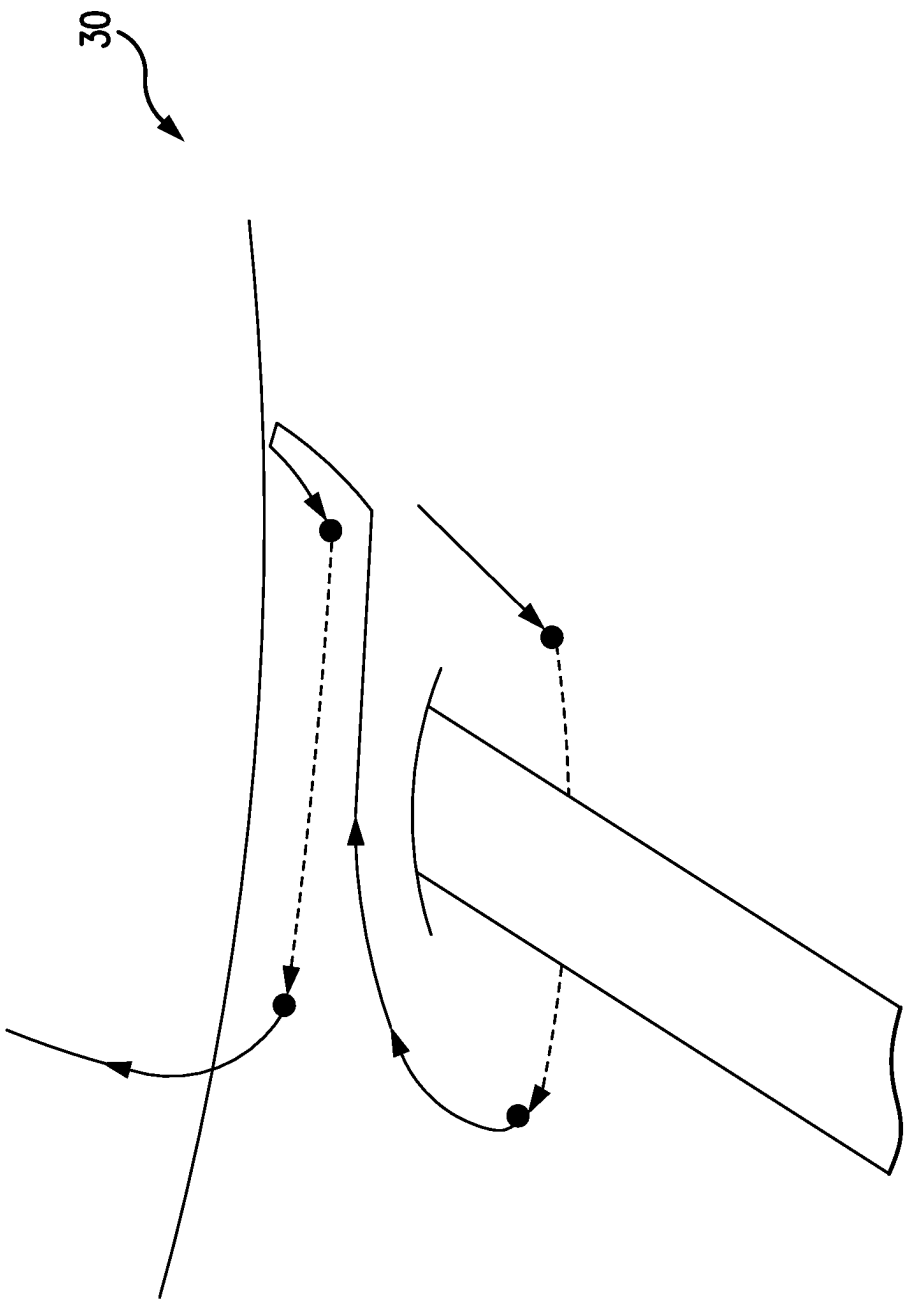


FIG. 8

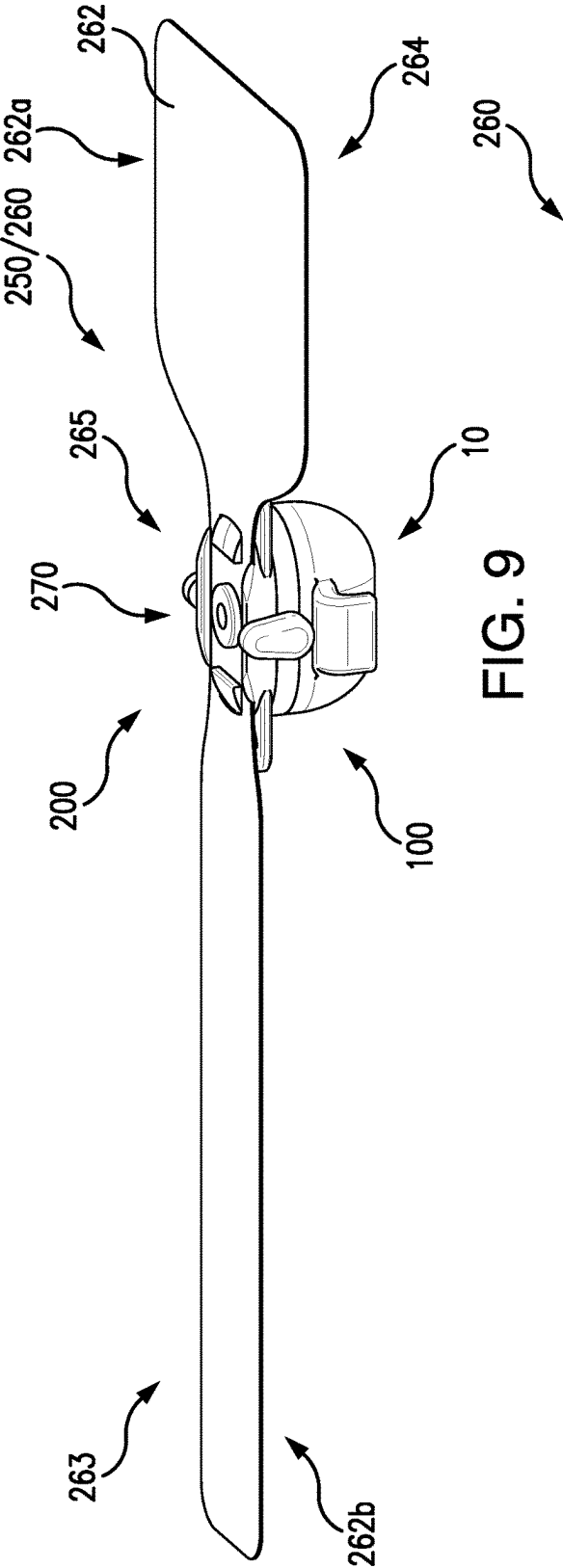


FIG. 9

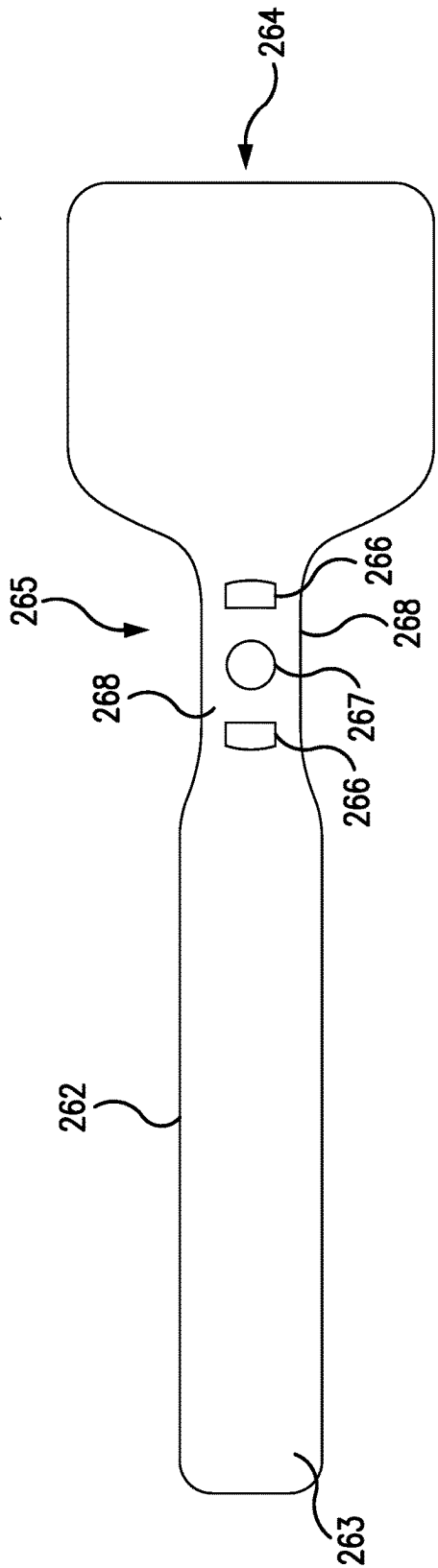


FIG. 10

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HEMOSTASIS AIDING DEVICE**CROSS-REFERENCE OF RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Application No. 63/236,437, filed Aug. 24, 2021, the entirety of which is incorporated by reference herein.

FIELD OF INVENTION

This invention generally relates to hemostasis aiding devices for use with wounds made in endovascular procedures.

BACKGROUND OF THE INVENTION

Many various procedures require vascular access, which creates a puncture wound that eventually will need to be closed. After the process is completed, steps must be taken to close the wound. Wound closure typically involves compression to control bleeding until hemostasis occurs. Ideally, wound closure serves to minimize blood loss, effect hemostasis, and render the patient ambulatory in a relatively short period of time.

Traditionally, wound closure has been a manual operation where a physician or trained clinician used manual hand pressure, using either one or two hands. This is the gold standard for wound closure. However, such a process is time consuming, requiring a physician or trained clinician to apply pressure for at least 30 minutes, while also requiring a patient lie completely flat for several hours. There are additional problems as well. For example, manual pressure that is too firm does not allow sufficient clotting factors to accumulate and can potentially occlude blood flow completely. Moreover, manual pressure could be applied unevenly: the person exerting manual pressure can tire, or the fingers may move or may not be placed properly. The person may also stop the application of pressure to examine the wound, causing a disruption of the maturing clot.

In response to these problems, various specialized vascular closure devices (VCDs) have been proposed. However, they have not fully solved the problems inherent in wound closures. Accordingly, there is an unfulfilled need in the art for a simple-to-use VCD which closely mimics “gold standard” manual wound closure.

SUMMARY OF THE INVENTION

The invention is directed at a hemostasis aiding device. In an aspect, the hemostasis aiding device includes a pressure applying system and a securing system. A main body can connect the pressure applying system to the securing system. The pressure applying system can include a malleable patient pressure subsystem and a non-malleable patient-pressure subsystem. In such aspects, the main body supports both the malleable and non-malleable patient pressure subsystem and the non-malleable subsystems.

In an aspect, the malleable patient pressure subsystem includes a silicone bulb portion. The silicone bulb portion can include a stabilization portion and a pressure point. In an aspect, the non-malleable patient pressure subsystem applies force to the pressure point of the silicone bulb.

The pressure applying system can include a suture securing subsystem and an external securing subsystem. The suture securing subsystem includes an interior channel to receive a suture and a suture locking mechanism to lock the

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hemostasis aiding device to the suture connected to a patient. The external securing subsystem can include an adhesive subsystem and a strap subsystem.

The accompanying drawings are included to provide a further understanding of the invention and are incorporated in and constitute part of this specification, as well as illustrate several embodiments of the invention that together with the description serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a schematic representation of a placement of an access sheath. Subcutaneous tissues contain small ovals for differentiation and the vasculature is lined with crosshatching.

FIG. 2 is a schematic representation of the hemostasis aiding device applied to the wound area of FIG. 1 according to an aspect of the present invention.

FIG. 3 is a perspective view of the hemostasis aiding device according to an aspect of the present invention.

FIG. 4 is a front plan view of the hemostasis aiding device of FIG. 3.

FIG. 5 is a top plan view of the hemostasis aiding device of FIG. 3.

FIG. 6 is a bottom plan view of the hemostasis aiding device of FIG. 3.

FIG. 7 is a cross-sectional view of the hemostasis aiding device of FIG. 3.

FIG. 8 is a schematic illustration of a suture utilized by a hemostasis aiding device according to an aspect of the present invention.

FIG. 9 is a perspective view of the hemostasis aiding device on a securing mechanism according to an aspect of the present invention.

FIG. 10 is a top plan view of the securing mechanism of FIG. 9.

DETAILED DESCRIPTION OF THE INVENTION

Embodiments of the invention will be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

In the following description, numerous specific details are set forth. However, it is to be understood that embodiments of the invention may be practiced without these specific details. In other instances, well-known methods, structures, and techniques have been shown in detail in order not to obscure an understanding of this description.

In an aspect, the invention is directed at a hemostasis aiding device 10, as shown in FIGS. 2-10. The hemostasis aiding device 10 is intended to aid in achieving hemostasis after a procedure that required femoral venous access, as shown in FIGS. 1-2. Similar/related procedures include catheter based cardiac ablation, transcatheter mitral valve replacement, catheter hemodialysis, and venous-arterial ECMO catheterizations. The device 10 works by applying pressure to the wound site 20 after the procedure. This site 20 is typically located in the groin crease. However, other areas of use include, but are not limited to, the arm. While

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it is possible to utilize the hemostasis aiding device **10** after procedures that required arterial and/or venous access, the device **10** is intended for use with venous applications given the lower level of blood pressure through veins in comparison to arteries.

Using this device **10** reduces post-operative recovery time, and also the bedside attendance of a caretaker, and improve patient comfort. That is, the device **10** applies the needed pressure to stop the bleeding without the need of additional pressure applying sources (e.g., clinician). In an aspect, the device **10** is equivalent to the gold standard of hemostasis, which is to have a clinician use manual closure and/or apply pressure until bleeding is stopped. Assuming complete device success, the gold standard intervention requires 10-20 minutes for first hemostasis and a repeat of that procedure/time if the clotting fails.

In an aspect, the device **10** is configured to be used as a single use sterile device **10**. However, in other aspects, components of the device **10** can be configured for multiple uses. As shown, the hemostasis aiding device **10** includes a main body **50** supporting a pressure applying system **100** and a securing system **200**. The pressure applying system **100** applies pressure directly to the wound area **20** to assist with hemostasis. The securing system **200** secures the pressure applying system **100** to the wound site **20**, but also assists in applying pressure via the pressure applying system **100**, as discussed below.

In an aspect, the main body **50** is integrated with both the pressure applying system **100** and the securing system **200**. In such instances, the main body **50** can share features of the applying system **100** and the securing system **200**. The main body **50** includes a top surface **52** and a bottom surface **54**. In an aspect, the pressure applying system **100** is coupled/formed on the bottom surface **54**, and the securing system **200** is coupled to/formed on portions of the top surface **52**. In an aspect, the main body **50** has a length **56** and a width **58**. In an aspect, the main body **50** can have an oblong shape, with the length **56** being greater than the width **58**. In such aspects, components of the pressure applying system **100** can share the general oblong shape, as discussed below. The main body **50** can also include an aperture **70**, discussed below.

The pressure applying system **100** can include a malleable portion **110** and a non-malleable portion **160**. The malleable portion **110** can include a malleable bulb **120**. In such aspects, the malleable bulb **120** is configured to be malleable, but capable of applying pressure to a wound area **20**. In an aspect, the malleable bulb **120** is made of a deformable material that is capable of transferring some force to the wound area without causing the subject pain or irritation. In an aspect, the deformable material is silicone, but in other aspects other deformable materials can be utilized. However, it is preferred that the material be capable of being sterilized as well as not cause skin irritation. Further, the malleable bulb **120** can be configured to be hollow, which aids in the deformity/malleability of the bulb **120**. Further, in such aspects, the bulb **120** being hollow provides a location for the non-malleable portion **150** to be housed, as discussed below.

The hollow malleable bulb **120** includes a top **122**, side walls **124** that extend from the top **122** to a bottom **126** that includes a pressure point **128** (discussed below), forming an exterior surface **130**, and an interior surface **132**. In an aspect, protrusions/tabs can be found along the top **122** of the interior surface **132** that correspond to matching securing pockets **60** found on the bottom of the main body **50**. This

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protrusion/securing pockets interface, couples, and retains the malleable bulb **120** to the main body **50**.

In an aspect, the top **122** of the bulb **120** matches the oblong profile shape of the main body **50**, with the side walls **124** extending towards the bottom **126** in a rounded fashion. The curved nature of the bottom **126** forms the pressure point **128** on the exterior surface **130** of the bulb **120** that is configured to engage the skin of the wound site **20** and apply pressure. Extending upwards from the interior surface **132** of the bulb **120** is a connecting member **134** that is configured to engage the non-malleable portion **150**, discussed below. In an aspect, the connecting member **134** includes an aperture/channel **136**.

Given that the malleable bulb **120** is malleable as pressure is applied, it is possible that the pressure point **128** could move from the desired location, or that the bulb **120** is deformed and decreases the amount of pressure applied at the wound site **20**. To prevent this, the malleable portion **110** includes a stabilization subsystem **150**. The stabilization subsystem **150** is configured to keep the pressure point **128** of the malleable bulb **120** at the right location at the wound site **20** and in an orientation that maintains the desired amount of pressure at the wound sight/pressure point interface.

In an aspect, the stabilization subsystem **150** includes a pair of stabilization wings **152**. These stabilization wings **152** can extend from the sidewalls **124** on the exterior surface **130** of the malleable bulb **120** at opposite positions. In such aspects, the stabilization wings **152** are malleable as well, and made from the same material as the malleable bulb **120** and integrated into the side walls **124**. However, in other aspects, the stabilization wings **152** can be detachable. In an aspect, the stabilization wings **152** include a horizontal portion **153** that extends from the side walls **124**, and a vertical portion **154** that extend downward from ends of the horizontal portions **153** opposite the side walls **124**. In an aspect, the vertical portions **154** include edges **155** that are configured to engage the skin of the subject adjacent the wound site **20**. As the stabilization wings **152** provide support while the malleable bulb **120** applies pressure to the wound pressure and becomes depressed, the edges **155** are found on a plane above the plane of the bottom **126**/pressure point **128** of the bulb **120**. In other words, the bulb **120** has to be depressed before the edges **155** engage the skin of the subject.

As discussed above, the malleable portion **110** works with the non-malleable pressure applying portion **160** to apply pressure to the wound. The non-malleable pressure applying portion **160** includes a base **162** that is attached to/integrated the bottom surface **54** of the main body **50**. In an aspect, the base **162** can house the securing pockets **60**. The base **162** has a length **164** and a width **166**. The shape of the base **162** can correspond to the shape of the main body **50**. For example, the base **162** can have an oblong shape that corresponds to the oblong shape of the main body, with the length **164** being greater than the width **166**.

A tapered member **170** can extend away from the base **162**. The tapered member **170** has a length **172**, width **174**, and a depth **176**, with the length **172** and the width **174** decreasing in the depth **176** direction. A bottom **178** is found at the end of the tapering member **170** opposite the base **162**. As discussed above, the non-malleable portion **160** can be housed within the hollow malleable bulb **120**, with the bottom **178** of the tapered member **170** engaging the interior protrusion **134** of the interior surface **130** of the bulb **120**. The non-malleable pressure applying portion **160** is configured to apply pressure only after the malleable portion **110**

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has begun to substantially apply pressure. In other words, the malleable bulb **120** has applied pressure and is being deformed before the bottom **178** of the tapered member **170** begins to apply pressure. In addition, the non-malleable portion **160** provides additional stabilization. The tapered member **170** can include an aperture **180** that extends through its body and is in alignment with the aperture **136** of the bulb **120**.

The pressure applying system **100** works with the securing system **200** to ensure that pressure is applied correctly to the wound site **20**. In an aspect, the securing system **200** can include a suture securing subsystem **210** and an external securing subsystem **250**. In an aspect, the suture securing subsystem **210** works with a suture **30** applied at the wound site **20**. For example, a Z-suture **30** as shown in FIG. **8** can be utilized. Various materials can be used for the suture **30**. In an aspect, a 2-0 Ethibond suture **30** on a CT-1 needle can be utilized. However, other sutures, materials and shape, can be utilized. In various types, the suture **30** should not be knotted, but secured by the device, as discussed below.

The suture securing subsystem **210** receives the suture **30** via an interior channel **212**. The interior channel **212** extends through multiple components of the device **10**, formed from the aperture **136** of the bulb **120**, the aperture **180** of the tapered member **170**, and the aperture **70** of the main body **50**. The suture securing subsystem **210** also includes a suture locking mechanism **220**. The suture locking mechanism **220** can be found within the interior of the main body **50**. The suture locking mechanism includes a rotating member **222** housed within a horizontal locking channel **230**. The rotating member **222** can include a spherical portion **224**. The spherical portion **224** includes an aperture **226** that forms part of the interior channel **212**. The locking channel **230** includes a corresponding spherical hollow portion **232** that houses the spherical portion **224** of the rotating member **222**. Knobs **228** can be found at the ends of the rotating member **222** which allows a physician or medical professional to rotate the rotating member **222**.

The suture securing subsystem **210** includes a mount extension **240** found on the top surface **52** of the main body **50**. The mount extension **240** is raised above the top surface **52** of the main body **50**, and includes a flange **242**. In an aspect, the flange **242** has a circular shape. The mount extension **240** includes an aperture **244** that is part of the interior channel **212**.

The suture securing subsystem **210** works with the external securing subsystem **250**. The external securing subsystem **250** can be configured to apply the needed force to help hemostasis at the wound site **20**. In an aspect, the external securing subsystem **250** can include an adhesive subsystem **260** and/or a strap subsystem **280**.

In an aspect, the adhesive subsystem **260** includes an adhesive strap **262** and a mounting component **270**. The adhesive strap **262** includes a top surface **262a** and an adhesive bottom surface **262b**. The adhesive strap **262** includes an elongated portion **263**, a fat portion **264**, and a mount portion **265** connecting the elongated portion **263** to the fat portion **264**. The mount portion **265** matches the shape of the mount component **270**. As shown, the mount portion **265** includes notch apertures **266** and a flange aperture **267**, and edges **268**, discussed below. In other aspects, the adhesive strap **262** can have portions of different size and shapes dependent on the location of the wound to be addressed. For example, the adhesive strap **262** as shown in FIGS. **9-10** is shaped accordingly to avoid a long strap on

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the inner side so it does not encroach on a middle part of the pelvic area when used on a wound associated with femoral venous access.

The mount component **270** is configured to receive the adhesive strap **262** at the mount portion **265**. The mount component **270** is found on top surface **52** of the main body **50** of the device **10**. The mount component **270** includes a combination of tabs, nooks, and side members to secure the adhesive strap **262**. As shown in FIG. **7**, the mount component **270** includes the mount extension **240** with its flange **242**, an inlet **272** with tabs **274** and side members **276**. The tabs **274** and side securing members **276** can both include notches **275**, **277**, respectively, that engage the notch apertures **266** and edges **268** of the adhesive strap **262**, with the flange **242** also securing the flange aperture **267**.

The strap subsystem **280** can include d-rings **282** found on the main body **50**. A Velcro strap (not shown) can be utilized with the d-rings **282** to secure the main body **50**, and the pressure system **100**, to the patient. In various aspects, the adhesive subsystem **260** and the strap subsystem **280** can come in various lengths and sizes to accommodate the different sizes of patients and limbs.

To apply the device **10**, a suture **30** (e.g., FIG. **8**) is made at the wound site **20**. The ends of the suture **30** are then fed through the interior channel **212**, starting at the aperture **136** of the bulb **120** and then pulled through the aperture **70** of the main body **50**. Once it is pulled through, the end of the suture **30** is then wrapped around the mount extension **240** and the suture locking mechanism **220** is engaged by turning the knobs **228**. As the knobs **228** are turned, the rotating member **222** is rotated. The edges of the aperture **226** of the spherical member **224** engage the suture **30**, and pull it tighter within the spherical hollow **232** of the horizontal channel **230**. This “locks” the device **10** on the suture **30** at the right location. In an aspect, the rotating member **222** rotates 90 degrees to facilitate a user friendly binary lock or unlock. However, in other aspects, the rotating member **222** can have an infinite rotation that allows the user to draw-up the suture **30** to provide extra tension.

From here, the external securing subsystem **250** is utilized which may utilize either the adhesive subsystem **260** or strap subsystem **280** to further secure the device **10** to the subject. If the adhesive subsystem **260** is utilized, the mount portion **265** of the adhesive strap **262** is placed within the mount component **270** of the main body **50**, with the tabs **274** being fed into the notch apertures **266**, with their notches engaging the notch aperture **266**, the flange **242** into the flange aperture **267**, and the edges **268** placed within the notches **277** of the side securing members **276**. From here, the elongated portion **263** and the fat portion **264** are tightened and adhered to the surface of the subject’s skin near the wound site **20**. The adhesive subsystem **260** can be pulled taught to apply more pressure. If the strap subsystem **280** is utilized, a Velcro strap is fed through the d-rings **282** and then tightened.

In either case, when the external securing subsystem **250** is engaged, pressure is applied to the device **10**, which activates the pressure applying system **100**, including both the malleable and non-malleable portions **110**, **160**. The pressure point **128** of the malleable bulb **120** is pressed against the wound surface. As the bulb **120** is deformed, the stabilizing subsystem **150** is engaged, with the wings **152** keeping the pressure point **128** in place. At the same time, the non-malleable portion **160** applies pressure to the pressure point **128**. The securing system **200** remains engaged until hemostasis occurs.

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The device **10** discussed above produces a force that substitutes for the needed force applied by a physician or nurse. For example, the pressure applied by the hemostasis aiding device **10** over a number of trials was between 8 and 14 Newtons with the true average being ~12 N, which is enough force to translate pressure through potentially thick tissue to the venous site. In addition, the device **10** can be applied fairly quickly. Device placement with the sheet adhesive took between 13 and 30 seconds (20 second average) without the suture. If a suture is being used (not including suture placement time) the time to placement ranged from 22-70 seconds with the average approximately 42 seconds.

Having thus described exemplary embodiments of the present invention, it should be noted by those skilled in the art that the disclosures are exemplary only and that various other alternatives, adaptations, and modifications may be made within the scope of the present invention. Accordingly, the present invention is not limited to the specific embodiments as illustrated herein, but is only limited by the following claims.

What is claimed is:

1. A hemostasis aiding device for use with a subject after a venous access procedure, the device comprising:
 - a. a securing system to secure the hemostasis aiding device proximate a wound site; and
 - b. a pressure applying system to apply pressure to the wound site of the venous access procedure of the subject, wherein after the hemostasis aiding device is secured to the subject, the pressure applying system is configured to apply pressure without additional pressure applying sources, wherein the pressure applying system includes:
 - i. a malleable portion configured to engage the wound site, the malleable portion including a pressure point to apply pressure at the wound site;
 - ii. a non-malleable portion configured to apply pressure to the malleable portion which applies pressure to the wound site; and
 - iii. a stabilization subsystem configured to keep the pressure point at the wound site as pressure is applied, wherein the stabilization subsystem comprises stabilization wings, wherein each of the stabilization wings includes a horizontal portion extending away from the pressure point, a vertical portion extending downwards from the horizontal portion, and an edge for engaging skin of the subject when pressure is applied.
2. A hemostasis aiding device for use with a subject after a venous access procedure, the device comprising:
 - a. a pressure applying system comprising:
 - i. a main body having an oval shape having a length and a width, wherein the main body is configured to be placed with the length perpendicular to a direction of a vein of the subject;
 - ii. a malleable patient pressure interface portion positioned below the main body comprising:
 - A. a silicone bulb portion having:
 - a. a bottom portion; and
 - b. a pressure point, wherein the bottom portion forms a rounded shape to create the pressure point that, when compressed, flexes and presses into skin;
 - B. a stabilization portion associated with the malleable patient pressure interface portion comprising;

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- a. a pair of stabilization wings, the wings oriented on opposite sides of the silicone bulb, each wing configured to engage the skin of the subject and keep the silicone bulb from rotating when pressure is applied to the hemostasis aiding device; and
 - iii. a non-malleable pressure applying portion positioned on a bottom surface of the main body and extending downward within the malleable patient pressure interface portion; and
 - b. a subject securing portion configured to secure the device to the patient comprising:
 - i. a suture securing component comprising:
 - A. a suture receiving portion; and
 - B. a suture locking mechanism; and
 - ii. an external securing component configured to work with the pressure applying system to apply additional force comprising:
 - A. an interface on the pressure applying system; and
 - B. a patient interface portion configured to be secured to the subject.
3. The hemostasis aiding device of claim 1, wherein the securing system is oriented above the pressure applying system when the pressure applying system is applying pressure at the wound site.
4. The hemostasis aiding device of claim 3, further comprising a main body having a top surface and a bottom surface, wherein the securing system is coupled to the top surface and the pressure applying system is coupled to the bottom surface.
5. The hemostasis aiding device of claim 4, wherein the main body and the pressure applying system have an oblong shape.
6. A hemostasis aiding device for use with a subject after a venous access procedure, the device comprising:
 - a. a securing system to secure the hemostasis aiding device proximate a wound site; and
 - b. a pressure applying system to apply pressure to the wound site of the venous access procedure of the subject, wherein after the hemostasis aiding device is secured to the subject, the pressure applying system applies pressure without additional pressure applying sources, wherein the pressure applying system includes:
 - i. a malleable portion comprising a silicone bulb; and
 - ii. a non-malleable portion comprises a tapered member housed within the malleable portion, wherein the malleable portion is configured to engage the wound site and the non-malleable portion applies pressure through the tapered member after the malleable portion applies pressure to the wound site.
7. The hemostasis aiding device of claim 6, wherein the malleable portion includes a pressure point to apply pressure to the wound site.
8. The hemostasis aiding device of claim 7, wherein the pressure applying system includes a stabilization subsystem configured to keep the pressure point at the wound site as pressure is applied.
9. The hemostasis aiding device of claim 8, wherein the stabilization subsystem comprises stabilization wings.
10. The hemostasis aiding device of claim 9, wherein each of the stabilization wings includes a horizontal portion extending away from the pressure point, a vertical portion extending downwards from the horizontal portion, and an edge for engaging skin of the subject when pressure is applied.

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11. A hemostasis aiding device for use with a subject after a venous access procedure, the device comprising:

- a. a securing system to secure the hemostasis aiding device proximate a wound site, the securing system comprising a suture securing subsystem for use with a suture secured at the wound site, the suture securing subsystem comprising:
 - i. an interior channel and a suture locking mechanism to receive the suture; and
- b. a pressure applying system to apply pressure to the wound site of the venous access procedure of the subject, wherein after the hemostasis aiding device is secured to the subject, the pressure applying system applies pressure without additional pressure applying sources.

12. The hemostasis aiding device of claim 11, wherein the suture locking mechanism includes a rotating member that receives the suture, wherein when the rotating member is rotated, the suture is locked in place within the suture securing subsystem.

13. The hemostasis aiding device of claim 1 or 6, wherein the securing system comprises a suture securing subsystem for use with a suture secured at the wound site.

14. The hemostasis aiding device of claim 13, wherein the suture securing subsystem comprises an interior channel and a suture locking mechanism to receive the suture.

15. The hemostasis aiding device of claim 14, wherein the suture locking mechanism includes a rotating member that

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receives the suture, wherein when the rotating member is rotated, the suture is locked in place within the suture securing subsystem.

16. The hemostasis aiding device of claim 1 or 6, wherein the securing system comprises an external securing subsystem configured to apply additional force.

17. The hemostasis aiding device of claim 16, wherein the external securing subsystem comprises an adhesive subsystem and a mounting component, the adhesive subsystem configured to adhere to the subject and to be received by the mounting component, wherein the mounting component is coupled to the pressure applying system.

18. The hemostasis aiding device of claim 1, 2, 11, or 6, wherein after being secured to the subject, the hemostasis aiding device is configured to apply a force of about 8 Newtons to about 14 Newtons.

19. The hemostasis aiding device of claim 1, wherein the wound site is located at a groin crease of the subject.

20. The hemostasis aiding device of claim 1, wherein the malleable portion comprises a silicone bulb.

21. The hemostasis aiding device of claim 20, wherein the non-malleable portion comprises a tapered member housed within the malleable portion, the tapered member applying pressure to the malleable portion after the malleable portion has applied pressure to the wound site.

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